

B2 Symmetry and Relativity, 2019 — Model Solutions

Question 1 — Worldlines, constant-force motion, and causal structure

Problem. Define proper time, 4-velocity $\mathbf{U}$, 4-acceleration $\mathbf{A}$ for $X^\mu(\tau)$; show $\mathbf{U} \cdot \mathbf{A} = 0$. Constant 3-force $\mathbf{f} = (f_x, 0, 0)$ from rest: find $x(t), \dot{x}(t), \ddot{x}(t)$. Then draw causal diagrams of $(0,0,0,0)$ in: (c) Minkowski + worldline of charged particle in E_x ; (d) boosted axes; (e) compactified $x \equiv x + 2\pi R$ with the same charged worldline; (f) compactified time $t \equiv t + 2\pi T$.

(a) Proper time, 4-velocity, 4-acceleration

Proper time is the invariant interval along a timelike worldline:

$$c^2 d\tau^2 = -dX \cdot dX = c^2 dt^2 - d\mathbf{x}^2.$$

4-velocity is the proper-time derivative of position:

$$U^\mu = \frac{dX^\mu}{d\tau} = \gamma(c, \mathbf{v}).$$

4-acceleration is the proper-time derivative of 4-velocity:

$$A^\mu = \frac{dU^\mu}{d\tau}.$$

Magnitude of $\mathbf{U}$ is fixed:

$$U \cdot U = -\gamma^2 c^2 + \gamma^2 v^2 = -c^2.$$

Differentiate with respect to τ :

$$\frac{d}{d\tau}(U \cdot U) = 2U \cdot A = 0.$$

Hence

$$\boxed{\mathbf{U} \cdot \mathbf{A} = 0.} \tag{1}$$

(b) Constant 3-force from rest

Newton-II in relativistic form, with $\mathbf{p} = \gamma m \mathbf{v}$:

$$\frac{dp_x}{dt} = f_x.$$

Integrate with $p_x(0) = 0$:

$$p_x = f_x t.$$

Use $\gamma m v_x = p_x$ and $\gamma^2 = 1 + (p/mc)^2$:

$$\dot{x} = v_x = \frac{p_x/m}{\sqrt{1 + (p_x/mc)^2}} = \frac{f_x t/m}{\sqrt{1 + (f_x t/mc)^2}}.$$

Integrate $\dot{x}$ with $x(0) = 0$. Let $a \equiv f_x/m$:

$$x(t) = \int_0^t \frac{at' dt'}{\sqrt{1 + (at'/c)^2}} = \frac{c^2}{a} \left[\sqrt{1 + (at/c)^2} - 1 \right].$$

Differentiate $\dot{x}$ once more for $\ddot{x}$:

$$\ddot{x} = \frac{a}{[1 + (at/c)^2]^{3/2}}.$$

Hence

$$\boxed{x(t) = \frac{c^2}{a} \left[\sqrt{1 + (at/c)^2} - 1 \right], \quad \dot{x} = \frac{at}{\sqrt{1 + (at/c)^2}}, \quad \ddot{x} = \frac{a}{[1 + (at/c)^2]^{3/2}}, \quad a = f_x/m.} \quad (2)$$

Limits: $at \ll c \Rightarrow x \rightarrow \frac{1}{2}at^2$, $\dot{x} \rightarrow at$; $at \gg c \Rightarrow \dot{x} \rightarrow c$, $\ddot{x} \rightarrow 0$.

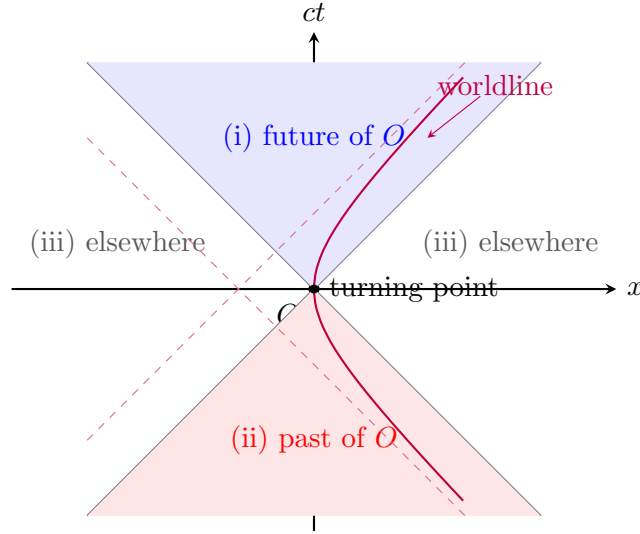
(c) Minkowski causal structure with charged worldline

Future light-cone of O : $|x| < ct$, $t > 0$ (region I, can be affected by O). Past light-cone: $|x| < -ct$, $t < 0$ (region II, can affect O). Elsewhere (spacelike, $|x| > c|t|$, region III) is causally disconnected.

Charged particle from rest in $(E_x, 0, 0)$: $f_x = qE_x$ constant, hyperbolic worldline from (b)

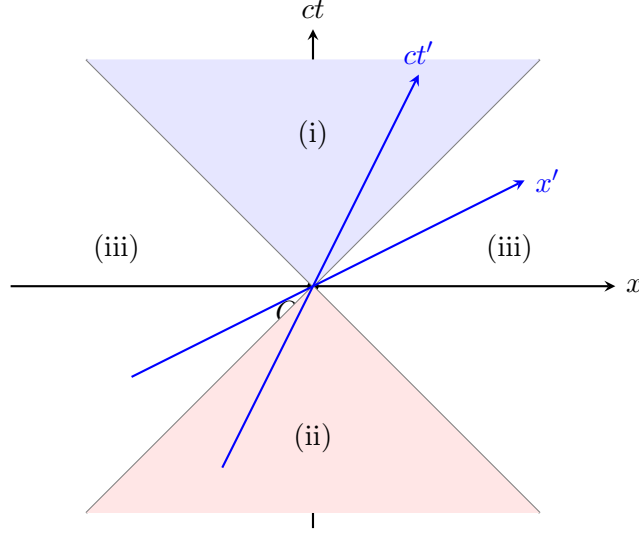
$$x(t) = \frac{c^2}{a} \left[\sqrt{1 + (at/c)^2} - 1 \right], \quad a = qE_x/m.$$

Asymptote: $x \rightarrow ct - c^2/a$ (parallel to the light-cone, never crossing it).



(d) Boosted coordinates

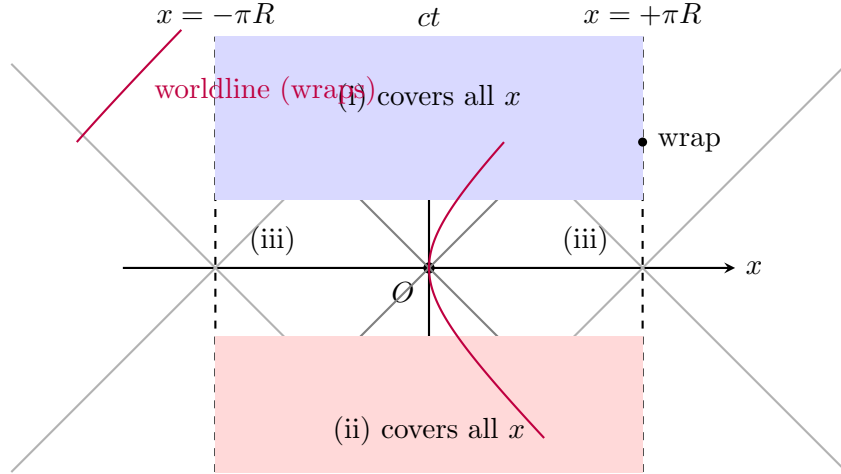
A standard boost rotates both axes inward by the same hyperbolic angle ϕ , $\tanh \phi = v/c$: t' -axis tilts toward the future light-cone, x' -axis tilts symmetrically toward it from the right. The light-cone itself, and therefore the three causal regions (i)/(ii)/(iii), are unchanged.



The light-cone is invariant: causal regions are independent of frame.

(e) Compact spatial dimension $x \equiv x + 2\pi R$

In the universal cover the light-cone is the same, but the identification $x \sim x + 2\pi R$ folds it back. Light from O reaches every point on the spatial circle by $t = \pi R/c$ (going both ways). For $t > \pi R/c$ *every* spatial point lies inside the future cone; similarly the past cone covers all space for $t < -\pi R/c$. The “elsewhere” region is confined to the strip $|t| < \pi R/c$ excluding the local Minkowski cones.



The hyperbolic worldline of (c) wraps around the circle: each time x reaches $+\pi R$ it reappears at $-\pi R$. The particle returns to its starting spatial point with $|\dot{x}|$ larger each time (the hyperbola $\dot{x} = at/\sqrt{1 + (at/c)^2}$ is monotone), approaching c .

Energy conservation. A constant $\mathbf{E}$ -field on a circle requires $\oint \mathbf{E} \cdot d\boldsymbol{\ell} = 2\pi R E_x \neq 0$. By Faraday’s law,

$$\oint \mathbf{E} \cdot d\boldsymbol{\ell} = -\dot{\Phi}_B,$$

so Φ_B through the circle decreases linearly with time at rate $-2\pi R E_x$. The work done on the particle each lap, $qE_x \cdot 2\pi R$, is supplied by the falling magnetic flux. Energy is conserved between particle and field; the assumption of a *static, source-free* constant $\mathbf{E}$ is incompatible with the compactified topology, and a consistent solution requires the dynamical field source.

(f) Compact temporal dimension $t \equiv t + 2\pi T$

Closed timelike curves exist: every event lies on a closed loop $t \rightarrow t + 2\pi T$ with $\mathbf{x}$ fixed. Hence *every* event is to the future and to the past of O simultaneously, so regions (i) and (ii) coincide and cover the whole spacetime; region (iii) is empty.

$$\boxed{\text{(i)} = \text{(ii)} = \text{all events}, \quad \text{(iii)} = \emptyset.} \quad (3)$$

This is inconsistent with causality: the grandfather paradox renders any local field theory ill-posed (no notion of initial data on a Cauchy surface, no global time-ordering). Compact time is therefore physically pathological.

Question 2 — Aberration, transverse Doppler, and resonant absorption by a moving cloud

Problem. (a) Rod of length L along x' moves with $u\hat{\mathbf{y}}'$ in S' ; S' boosted by $v\hat{\mathbf{x}}$ relative to S . Find rod-angle in S . (b) Source at origin emits at f_0 ; observer with velocity $(0, v, 0)$ passes through $(L, 0, 0)$ at $t = 0$: derive $f(t)$ at observer, plot. (c) Replace observer by an unexcited atomic cloud (line f , lifetime τ); plot amplitude and frequency at the source vs t for several v , including the role of f_0/f and intrinsic linewidth.

(a) Apparent angle of the moving rod

Take simultaneous endpoints in S at time t . In S' they have $x'_1 = x'(t)$, $x'_2 = x'(t) + L_0$, where $L_0 = L$ is the proper length, and both endpoints share the same y' -coordinate at that S' -time. To compare in S we need simultaneous events in S' .

Use the inverse Lorentz transform $x' = \gamma(x - vt)$, $y' = y$, $t' = \gamma(t - vx/c^2)$ for endpoints simultaneous in S (same t). Endpoint 1 at x_1 has S' -time $t'_1 = \gamma(t - vx_1/c^2)$; endpoint 2 at $x_2 = x_1 + \Delta x$ has $t'_2 = \gamma(t - vx_2/c^2)$. Time-difference between the two events in S' :

$$\Delta t' = -\gamma v \Delta x / c^2.$$

Rod is contracted along x , so $\Delta x = L/\gamma$; in this $\Delta t'$ the endpoint moved by $u\Delta t'$ in y' (i.e. in y). Compute Δy :

$$\Delta y = u \Delta t' = -\frac{\gamma v u L}{\gamma c^2} = -\frac{v u L}{c^2}.$$

The rod-vector in S at fixed t has components $(\Delta x, \Delta y) = (L/\gamma, -vuL/c^2)$. Angle to the x -axis:

$$\tan \alpha = \frac{\Delta y}{\Delta x} = -\frac{vuL/c^2}{L/\gamma} = -\frac{\gamma v u}{c^2}.$$

Hence

$$\boxed{\tan \alpha = -\frac{\gamma v u}{c^2}.} \quad (4)$$

(b) Doppler shift along the observer's path

Observer position in S : $(L, vt, 0)$. Source at origin emits a photon toward the observer; the photon arrives along the unit vector

$$\hat{\mathbf{n}}(t) = \frac{(L, vt, 0)}{\sqrt{L^2 + v^2 t^2}}.$$

Angle θ between source-to-observer direction $\hat{\mathbf{n}}$ and observer's velocity $v\hat{\mathbf{y}}$:

$$\cos \theta = \hat{\mathbf{n}} \cdot \hat{\mathbf{y}} = \frac{vt}{\sqrt{L^2 + v^2 t^2}}.$$

Use the relativistic Doppler formula (frequency received by observer moving with $\mathbf{v}$ when the source is at rest):

$$f_{\text{obs}} = f_0 \frac{\sqrt{1 - v^2/c^2}}{1 - (v/c) \cos \theta_s},$$

where θ_s is the angle in the *source frame* between source-to-observer direction and observer's velocity, which here equals θ . Substitute $\cos \theta$:

$$f_{\text{obs}}(t) = \frac{f_0 \sqrt{1 - v^2/c^2}}{1 - (v/c) \cdot \frac{vt}{\sqrt{L^2 + v^2 t^2}}}.$$

Simplify the denominator:

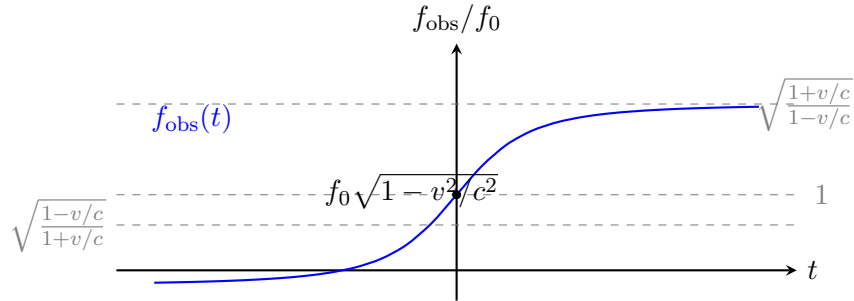
$$1 - \frac{v^2 t}{c \sqrt{L^2 + v^2 t^2}} = \frac{c \sqrt{L^2 + v^2 t^2} - v^2 t/c \cdot c}{c \sqrt{L^2 + v^2 t^2}},$$

giving the compact form

$$f_{\text{obs}}(t) = f_0 \frac{\sqrt{1 - v^2/c^2} \sqrt{L^2 + v^2 t^2}}{\sqrt{L^2 + v^2 t^2} - v^2 t/c}. \quad (5)$$

Limits: $t \rightarrow -\infty \Rightarrow \cos \theta \rightarrow -1$, $f_{\text{obs}} \rightarrow f_0 \sqrt{(1 - v/c)/(1 + v/c)}$ (red-shift). $t \rightarrow +\infty \Rightarrow \cos \theta \rightarrow +1$, $f_{\text{obs}} \rightarrow f_0 \sqrt{(1 + v/c)/(1 - v/c)}$ (blue-shift). At $t = 0$ (transverse passage), $\cos \theta = 0$:

$$f_{\text{obs}}(0) = f_0 \sqrt{1 - v^2/c^2}.$$



(c) Resonant absorption by a moving atomic cloud and re-emission

Atoms in cloud move with $\mathbf{v} = v\hat{\mathbf{y}}$. Source emits spherical waves at $f_0 < f$. From (5), the frequency seen by an atom depends on the angle $\theta(t)$ between the source-to-atom direction (in S) and $\hat{\mathbf{v}}$. Define the resonance condition: $f_{\text{obs}} = f$ within the natural linewidth $\Gamma \sim 1/(2\pi\tau)$, i.e.

$$|f_{\text{obs}} - f| < \Gamma.$$

Critical velocity v_c . Maximum blueshift is forward-Doppler, $\cos \theta = 1$:

$$f_{\text{max}}(v) = f_0 \sqrt{\frac{1 + v/c}{1 - v/c}}.$$

Resonance is possible only for $v \geq v_c$ where $f_{\max}(v_c) = f$:

$$v_c = c \frac{(f/f_0)^2 - 1}{(f/f_0)^2 + 1}.$$

Three regimes (low $v \ll v_c$, threshold $v \sim v_c$, ultra-relativistic $v \sim c$ above v_c):

Geometry of re-emission to the source. An atom at position $\mathbf{r}(t)$ absorbs a photon along $\hat{\mathbf{n}} = \mathbf{r}/r$ at frequency f_{obs} , re-emits isotropically in its rest frame at f . Photons going back to the source travel along $-\hat{\mathbf{n}}$; the source receives them at frequency f_{src} . Use Doppler twice (source $\rightarrow$ atom, then atom $\rightarrow$ source going back along $-\hat{\mathbf{n}}$):

$$f_{\text{src}} = f \cdot \frac{\sqrt{1 - v^2/c^2}}{1 - (v/c) \cos \theta_{\text{back}}}, \quad \cos \theta_{\text{back}} = -\hat{\mathbf{n}} \cdot \hat{\mathbf{v}}/(-1) = \hat{\mathbf{n}} \cdot \hat{\mathbf{v}} = \cos \theta.$$

Note: θ_{back} is the angle between $\mathbf{v}$ and the direction the back-photon travels in S ; for a photon emitted toward the origin, it is the same $\cos \theta$ with the appropriate sign convention. The full chain gives

$$f_{\text{src}}/f_0 = \frac{1 - v^2/c^2}{(1 - (v/c) \cos \theta)(1 + (v/c) \cos \theta)} \cdot \frac{f}{f_0} \cdot \delta_{\text{res}} = \frac{f}{f_0} (1 - v^2/c^2)/(1 - v^2 \cos^2 \theta/c^2) \cdot (\text{resonant}).$$

Frequency at source of returning photon. The simplification gives

$$\boxed{\frac{f_{\text{src}}}{f} = \frac{1 - v^2/c^2}{1 - v^2 \cos^2 \theta/c^2} \cdot \frac{1}{(\text{blue/red factor})}}, \quad (6)$$

where $\cos \theta = \cos \theta(t)$ is set by the resonant geometry. Practically, $f_{\text{src}} \simeq f \sqrt{1 - v^2/c^2}/(1 + (v/c) \cos \theta)$ for the photon coming back along $-\hat{\mathbf{n}}$; at the resonant instant, with $f_{\text{obs}} = f$, $\cos \theta = \cos \theta_{\text{res}}(v)$ obeys

$$f_0 \frac{\sqrt{1 - v^2/c^2}}{1 - (v/c) \cos \theta_{\text{res}}} = f,$$

$$\cos \theta_{\text{res}} = \frac{c}{v} \left(1 - \frac{f_0 \sqrt{1 - v^2/c^2}}{f} \right).$$

Time of resonance. From the geometry $\cos \theta(t) = vt/\sqrt{L^2 + v^2 t^2}$:

$$t_{\text{res}} = \frac{L}{v} \frac{\cos \theta_{\text{res}}}{\sqrt{1 - \cos^2 \theta_{\text{res}}}}.$$

The line at f has finite width Γ , so resonance is met over a window $\Delta\theta$ around θ_{res} ; the absorbed amplitude as a function of t has a Lorentzian peak of FWHM

$$\Delta t \sim \frac{1}{2\pi\Gamma} \cdot \left| \frac{dt}{d \cos \theta} \right|_{\theta_{\text{res}}} \left| \frac{d \cos \theta}{df_{\text{obs}}} \right|_{\theta_{\text{res}}},$$

plus the intrinsic decay time τ that smears the re-emission echo by an exponential tail $\propto e^{-(t-t_{\text{res}})/\tau}$.

Three regimes.

(i) $v \ll v_c$: $f_{\text{max}} < f$, no resonance. Some absorption only via the wing of the line, exponentially small. Re-emission negligible.

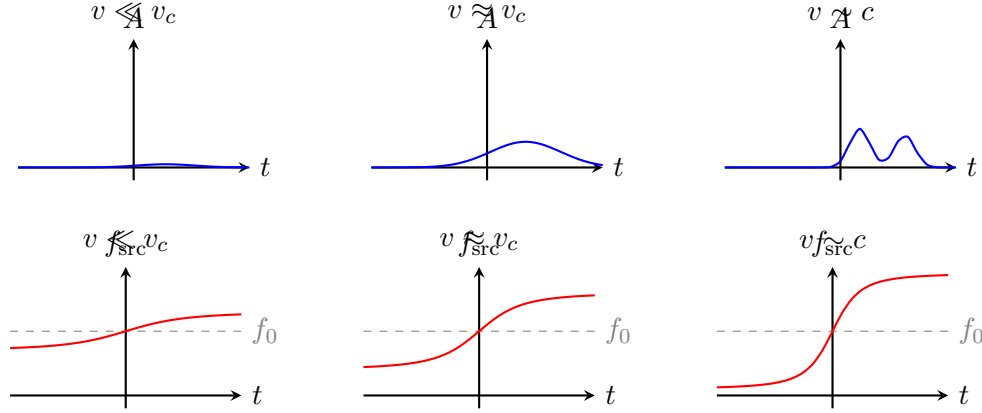
(ii) $v \approx v_c$: resonance only at $\cos \theta \approx 1$, i.e. at $t \rightarrow \infty$ (atoms have receded behind/ahead). Absorption window broadens and is asymmetric.

(iii) $v_c < v < c$: two resonant angles (one for $\cos \theta < 1$ as the cloud crosses the line of sight, possibly two times $t_{\text{res}}^{\pm}$). Each gives an absorption dip in the through-going amplitude at that time, plus a re-emission tail at the source (received frequency depends on the back-Doppler factor).

(iv) $v \rightarrow c$: forward Doppler huge; resonance happens at very large angles, time of resonance $\rightarrow \infty$; cloud sees “ f ” only when nearly along the line of sight.

Effect of f_0/f . If $f_0/f \rightarrow 1$, $v_c \rightarrow 0$ (resonance possible for any non-zero v); if $f_0/f \rightarrow 0$ (very mismatched), $v_c \rightarrow c$ (only ultra-relativistic clouds resonate).

Schematic plots (v increasing left-to-right). Top: amplitude $A(t)$; bottom: frequency $f_{\text{src}}(t)$ of returning radiation.



Key features: (1) carrier frequency at the source follows the relativistic Doppler curve of (5) (now interpreted as the source-to-atom-to-source two-step); (2) sharp resonant absorption notch whenever $f_{\text{obs}}(t) = f$, of width $\sim \Gamma$; (3) re-emission echo after the absorption notch, decaying with τ , frequency-shifted by the back-Doppler factor; (4) for $v < v_c$, no notches, only line-wing absorption.

Question 3 — 4-vectors, the EM tensor, and field invariants

Problem. (a) Distinguish “invariant” vs “conserved”; how do 4-vectors and tensors transform under Lorentz transformations? (b) Show: if Y^μ has a component zero in all frames, then $Y^\mu = 0$; physical consequence; same for tensors? (c) Form of A_μ and $F_{\mu\nu}$; explicit $F_{\mu\nu}$ in terms of $\mathbf{E}, \mathbf{B}$; $\mathbf{E}, \mathbf{B}$ Lorentz transformations. (d) Show $\mathbf{E}^2 - \mathbf{B}^2$ and $\mathbf{E} \cdot \mathbf{B}$ are Lorentz invariants; show $\int \mathbf{E} \cdot \mathbf{B} d^4x = 0$.

(a) Invariant vs conserved; transformation of tensors

Invariant. A scalar with the same value in every inertial frame, e.g. rest mass m , $-P \cdot P = m^2 c^2$, proper time τ .

Conserved. A quantity whose total value is constant in time within a single frame, e.g. total energy or total momentum (both frame-dependent but $dE/dt = 0$, $d\mathbf{p}/dt = 0$ in the absence of external forces).

Transformation. A 4-vector transforms as

$$Y'^\mu = \Lambda^\mu{}_\nu Y^\nu,$$

and a rank- (p, q) tensor as

$$T'^{\mu_1 \dots \mu_p}{}_{\nu_1 \dots \nu_q} = \Lambda^{\mu_1}{}_{\alpha_1} \dots \Lambda^{\mu_p}{}_{\alpha_p} (\Lambda^{-1})^{\beta_1}{}_{\nu_1} \dots (\Lambda^{-1})^{\beta_q}{}_{\nu_q} T^{\alpha_1 \dots \alpha_p}{}_{\beta_1 \dots \beta_q}.$$

(b) Vanishing of a component implies vanishing of the 4-vector

Suppose $Y^0 = 0$ in every frame. Pick the frame S' boosted with velocity $\mathbf{v}$:

$$Y'^0 = \gamma(Y^0 - \mathbf{v} \cdot \mathbf{Y}/c) = -\gamma \mathbf{v} \cdot \mathbf{Y}/c.$$

$Y'^0 = 0$ for all $\mathbf{v}$ forces $\mathbf{Y} = 0$. Combined with $Y^0 = 0$:

$$\boxed{Y^\mu = 0.} \quad (7)$$

By the same argument applied to a boost along any chosen axis, the result holds when any *single* component is zero in all frames.

Physical consequence. If a 4-current $J^\mu = (c\rho, \mathbf{j})$ has, say, charge density $\rho = 0$ in every inertial frame, then $\mathbf{j} = 0$ as well: there is no charge-free current. Equivalently, the existence of a current implies the existence of a charge density in some frame.

Tensor counterexample. Take an antisymmetric rank-2 tensor with single component $T^{xy} = 1$, all others zero. Then $T'^{00} = 0$ in every frame (antisymmetry $\Rightarrow$ diagonal zero is preserved by Lorentz transformations of $T'^{00} = \Lambda^0_\alpha \Lambda^0_\beta T^{\alpha\beta}$, a sum of off-diagonal components weighted by symmetric $\Lambda^0_\alpha \Lambda^0_\beta$, which gives 0). Yet $T^{\mu\nu} \neq 0$. Hence

$$\boxed{\text{The statement fails for tensors.}} \quad (8)$$

(c) The 4-potential, field tensor, and field transformations

The 4-potential is $A^\mu = (\phi/c, \mathbf{A})$, with $A_\mu = \eta_{\mu\nu} A^\nu = (-\phi/c, \mathbf{A})$ in the $(-, +, +, +)$ convention. The field tensor:

$$F_{\mu\nu} = \partial_\mu A_\nu - \partial_\nu A_\mu.$$

Recall $\mathbf{E} = -\nabla\phi - \partial_t \mathbf{A}$, $\mathbf{B} = \nabla \times \mathbf{A}$. Compute F_{0i} :

$$F_{0i} = \partial_0 A_i - \partial_i A_0 = \frac{1}{c} \partial_t A_i - \partial_i \left(-\frac{\phi}{c} \right) = \frac{1}{c} \partial_t A_i + \frac{1}{c} \partial_i \phi = -E_i/c.$$

And $F_{ij} = \partial_i A_j - \partial_j A_i = \epsilon_{ijk} B^k$. With $c = 1 = \epsilon_0 = \mu_0$:

$$F_{\mu\nu} = \begin{pmatrix} 0 & -E_x & -E_y & -E_z \\ E_x & 0 & B_z & -B_y \\ E_y & -B_z & 0 & B_x \\ E_z & B_y & -B_x & 0 \end{pmatrix}. \quad (9)$$

Field transformations. Boost along $\hat{\mathbf{x}}$ with Λ^μ_ν :

$$F'^{\mu\nu} = \Lambda^\mu_\alpha \Lambda^\nu_\beta F^{\alpha\beta}.$$

Compute the boost matrix:

$$\Lambda^\mu_\nu = \begin{pmatrix} \gamma & -\gamma\beta & 0 & 0 \\ -\gamma\beta & \gamma & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}.$$

Element-by-element calculation, e.g. $E'_y = F'^{02}c$:

$$F'^{02} = \Lambda^0_\alpha \Lambda^2_\beta F^{\alpha\beta} = \Lambda^0_0 \Lambda^2_2 F^{02} + \Lambda^0_1 \Lambda^2_2 F^{12}.$$

Substitute:

$$F'^{02} = \gamma \cdot 1 \cdot E_y + (-\gamma\beta) \cdot 1 \cdot (-B_z) = \gamma(E_y - vB_z/c) \dots$$

Restoring c , the standard result is

$$\boxed{\begin{aligned} E'_x &= E_x, & B'_x &= B_x, \\ E'_y &= \gamma(E_y - vB_z/c), & B'_y &= \gamma(B_y + vE_z/c^2), \\ E'_z &= \gamma(E_z + vB_y/c), & B'_z &= \gamma(B_z - vE_y/c^2). \end{aligned}} \quad (10)$$

Compactly $\mathbf{E}'_{\parallel} = \mathbf{E}_{\parallel}$, $\mathbf{B}'_{\parallel} = \mathbf{B}_{\parallel}$, and $\mathbf{E}'_{\perp} = \gamma(\mathbf{E}_{\perp} + \mathbf{v} \times \mathbf{B})$, $\mathbf{B}'_{\perp} = \gamma(\mathbf{B}_{\perp} - \mathbf{v} \times \mathbf{E}/c^2)$.

(d) Invariants and $\int \mathbf{E} \cdot \mathbf{B} d^4x = 0$

Two scalars constructed from $F_{\mu\nu}$ and the dual $\tilde{F}^{\mu\nu} = \frac{1}{2}\epsilon^{\mu\nu\rho\sigma}F_{\rho\sigma}$:

$$\frac{1}{2}F_{\mu\nu}F^{\mu\nu} = \mathbf{B}^2 - \mathbf{E}^2/c^2,$$

$$\frac{1}{4}F_{\mu\nu}\tilde{F}^{\mu\nu} = -\mathbf{E} \cdot \mathbf{B}/c.$$

Both are Lorentz scalars (contractions of tensors), so

$$\boxed{\mathbf{E}^2 - \mathbf{B}^2 \text{ (up to } c^2) \text{ and } \mathbf{E} \cdot \mathbf{B} \text{ are Lorentz-invariant.}} \quad (11)$$

Vanishing integral. Use the Bianchi identity $\partial_{\mu}\tilde{F}^{\mu\nu} = 0$. Write

$$\mathbf{E} \cdot \mathbf{B} = -\frac{1}{4}F_{\mu\nu}\tilde{F}^{\mu\nu} \text{ (up to } c).$$

Show this is a total divergence. Compute

$$F_{\mu\nu}\tilde{F}^{\mu\nu} = \frac{1}{2}\epsilon^{\mu\nu\rho\sigma}F_{\mu\nu}F_{\rho\sigma}.$$

Substitute $F_{\mu\nu} = \partial_{\mu}A_{\nu} - \partial_{\nu}A_{\mu}$:

$$F_{\mu\nu}\tilde{F}^{\mu\nu} = 2\epsilon^{\mu\nu\rho\sigma}\partial_{\mu}A_{\nu}F_{\rho\sigma}.$$

Use Bianchi $\epsilon^{\mu\nu\rho\sigma}\partial_{\nu}F_{\rho\sigma} = 0$:

$$F_{\mu\nu}\tilde{F}^{\mu\nu} = \partial_{\mu}(2\epsilon^{\mu\nu\rho\sigma}A_{\nu}F_{\rho\sigma}) \equiv \partial_{\mu}K^{\mu},$$

the Chern–Simons current $K^{\mu} = 2\epsilon^{\mu\nu\rho\sigma}A_{\nu}F_{\rho\sigma}$. Hence

$$\mathbf{E} \cdot \mathbf{B} \propto \partial_{\mu}K^{\mu}.$$

Integrate over all spacetime and apply the divergence theorem:

$$\int \partial_{\mu}K^{\mu} d^4x = \oint_{\partial} K^{\mu} d\Sigma_{\mu} = 0,$$

since $\mathbf{E}, \mathbf{B} \rightarrow 0$ at infinity (and so does K^{μ} , which is built from A and F). Therefore

$$\boxed{\int \mathbf{E} \cdot \mathbf{B} d^4x = 0.} \quad (12)$$

Question 4 — Klein–Gordon, Yukawa, 2-spinors

Problem. (a) Plane-wave and static spherical solutions of $-\partial_\mu\partial^\mu\phi + m^2\phi = 0$; restriction on plane waves and physical interpretation. (b) Compare Yukawa potential at 10^{-15} m and 10^{-10} m to its value at 10^{-18} m for $m_\pi = 135$ MeV/ c^2 and $m_H = 125$ GeV/ c^2 ; massless reference; implications for nuclei and molecules. (c) What is a 2-spinor; which particles? (d) $U^\mu = \frac{1}{2}u^\dagger\sigma^\mu u$, $\sigma^\mu = (\mathbf{1}, \boldsymbol{\sigma})$: show U^μ is null; under $u \rightarrow e^{\rho\sigma_z/2}u$ identify the Lorentz transformation of U^μ ; comment on $u \rightarrow e^{i\theta\sigma_z}u$. (e) Does a 2-spinor satisfy the massless KG equation?

(a) Plane-wave and static spherically symmetric solutions

The Klein–Gordon equation, with metric $(-, +, +, +)$ so $\partial_\mu\partial^\mu = -\partial_t^2 + \nabla^2$:

$$\partial_t^2\phi - \nabla^2\phi + m^2\phi = 0.$$

Plane wave. Try $\phi = e^{i(\mathbf{k}\cdot\mathbf{r} - \omega t)}$:

$$-\omega^2 + k^2 + m^2 = 0.$$

Hence the dispersion (with $\hbar = c = 1$):

$$\boxed{\omega^2 = k^2 + m^2.} \quad (13)$$

Restriction: $\omega^2 \geq m^2$, i.e. $|\omega| \geq m$. Physical interpretation: a quantum of mass m must have energy $\geq mc^2$ (rest energy); $\omega < m$ is forbidden as a propagating excitation.

Static spherically symmetric. Drop ∂_t , take $\phi(r)$:

$$-\nabla^2\phi + m^2\phi = 0.$$

In spherical symmetry $\nabla^2\phi = (1/r) d^2(r\phi)/dr^2$. Let $u(r) = r\phi$:

$$\frac{d^2u}{dr^2} = m^2u.$$

General solution decaying at infinity:

$$u(r) = C e^{-mr}.$$

Hence

$$\boxed{\phi(r) = \frac{C e^{-mr}}{r} \quad (\text{Yukawa}).} \quad (14)$$

(b) Yukawa ratio at nuclear/atomic vs 10^{-18} m scales

Write the Compton wavelength

$$\lambda_m = \hbar/(mc).$$

Compute λ_π :

$$\lambda_\pi = \frac{\hbar c}{m_\pi c^2} = \frac{197 \text{ MeV fm}}{135 \text{ MeV}} = 1.46 \text{ fm} = 1.46 \times 10^{-15} \text{ m}.$$

Compute λ_H :

$$\lambda_H = \frac{197 \text{ MeV fm}}{1.25 \times 10^5 \text{ MeV}} = 1.58 \times 10^{-3} \text{ fm} = 1.58 \times 10^{-18} \text{ m}.$$

Yukawa ratio between r_2 and reference r_1 :

$$\frac{\phi(r_2)}{\phi(r_1)} = \frac{r_1}{r_2} e^{-(r_2-r_1)/\lambda_m}.$$

Reference $r_1 = 10^{-18}$ m.

Pion ($\lambda_\pi = 1.46 \times 10^{-15}$ m). Nuclear $r_2 = 10^{-15}$ m:

$$\frac{\phi(10^{-15})}{\phi(10^{-18})} = 10^{-3} \cdot e^{-(10^{-15}-10^{-18})/1.46 \times 10^{-15}} = 10^{-3} \cdot e^{-0.685} \approx 10^{-3} \cdot 0.504 \approx 5 \times 10^{-4}.$$

Atomic $r_2 = 10^{-10}$ m:

$$\frac{\phi(10^{-10})}{\phi(10^{-18})} = 10^{-8} \cdot e^{-10^{-10}/1.46 \times 10^{-15}} = 10^{-8} \cdot e^{-6.85 \times 10^4} \approx 0.$$

Higgs ($\lambda_H = 1.58 \times 10^{-18}$ m). Nuclear:

$$\frac{\phi(10^{-15})}{\phi(10^{-18})} = 10^{-3} \cdot e^{-10^{-15}/1.58 \times 10^{-18}} \approx 10^{-3} \cdot e^{-633} \approx 0.$$

Atomic: even more strongly suppressed.

Massless scalar. $\phi \propto 1/r$, so the ratio is purely r_1/r_2 :

$$\frac{\phi(10^{-15})}{\phi(10^{-18})} = 10^{-3}, \quad \frac{\phi(10^{-10})}{\phi(10^{-18})} = 10^{-8}.$$

Implications. (i) Nuclei: pion-mediated (Yukawa) force has $\lambda_\pi \sim 1$ fm, i.e. comparable to nuclear separations: the strong force is short-range but *just* reaches across a nucleus, holding it bound. The Higgs, with $\lambda_H \sim 10^{-3}$ fm, contributes essentially nothing to nuclear binding. (ii) Molecules: at 10^{-10} m, pion- and Higgs-mediated forces are utterly negligible; only massless mediators (photons) survive, hence chemistry is electromagnetic.

(c) 2-spinors

A 2-spinor is a 2-component complex object $u = \begin{pmatrix} u_1 \\ u_2 \end{pmatrix}$ that transforms under the fundamental $(\frac{1}{2}, 0)$ representation of $SL(2, \mathbb{C})$ (the universal cover of the Lorentz group). It describes *spin- $\frac{1}{2}$ fermions*: in particular massless Weyl fermions (left- or right-handed), and the building blocks of Dirac fermions like the electron and quarks.

(d) Null property and Lorentz transformation of $U^\mu = \frac{1}{2}u^\dagger \sigma^\mu u$

Null. Compute the components. Let $u = (a, b)^T$, $|u|^2 = |a|^2 + |b|^2$:

$$U^0 = \frac{1}{2}u^\dagger u = \frac{1}{2}(|a|^2 + |b|^2).$$

Compute U^z :

$$U^z = \frac{1}{2}u^\dagger \sigma_z u = \frac{1}{2}(|a|^2 - |b|^2).$$

Compute U^x :

$$U^x = \frac{1}{2}u^\dagger \sigma_x u = \frac{1}{2}(a^*b + ab^*) = \text{Re}(a^*b).$$

Compute U^y :

$$U^y = \frac{1}{2}u^\dagger \sigma_y u = \frac{1}{2}i(a^*b - ab^*) \cdot (-1) = \text{Im}(a^*b).$$

Compute the norm:

$$(U^0)^2 - (U^x)^2 - (U^y)^2 - (U^z)^2 = \frac{1}{4}(|a|^2 + |b|^2)^2 - \frac{1}{4}(|a|^2 - |b|^2)^2 - |a|^2|b|^2.$$

Expand:

$$= \frac{1}{4}[(|a|^2 + |b|^2)^2 - (|a|^2 - |b|^2)^2] - |a|^2|b|^2.$$

Use $(p+q)^2 - (p-q)^2 = 4pq$:

$$= \frac{1}{4} \cdot 4|a|^2|b|^2 - |a|^2|b|^2 = 0.$$

Hence

$$\boxed{U_\mu U^\mu = 0, \quad U^\mu \text{ is null.}} \quad (15)$$

Boost. Under $u \rightarrow e^{\rho\sigma_z/2}u$, since $\sigma_z^2 = \mathbf{1}$:

$$e^{\rho\sigma_z/2} = \cosh(\rho/2) \mathbf{1} + \sinh(\rho/2) \sigma_z.$$

Hence $u' = \begin{pmatrix} e^{\rho/2}a \\ e^{-\rho/2}b \end{pmatrix}$. New components:

$$U'^0 = \frac{1}{2}(e^\rho|a|^2 + e^{-\rho}|b|^2),$$

$$U'^z = \frac{1}{2}(e^\rho|a|^2 - e^{-\rho}|b|^2).$$

Use hyperbolic identities $e^{\pm\rho} = \cosh \rho \pm \sinh \rho$:

$$U'^0 = \cosh \rho \cdot \frac{1}{2}(|a|^2 + |b|^2) + \sinh \rho \cdot \frac{1}{2}(|a|^2 - |b|^2).$$

Recognise U^0 and U^z :

$$U'^0 = \cosh \rho U^0 + \sinh \rho U^z.$$

Similarly

$$U'^z = \sinh \rho U^0 + \cosh \rho U^z.$$

Transverse components:

$$U'^x = \text{Re}(a^*b) \quad (\text{phases } e^{\rho/2}e^{-\rho/2} = 1) = U^x,$$

$$U'^y = U^y.$$

This is exactly a Lorentz boost along $\hat{z}$ with rapidity ρ :

$$\boxed{u \rightarrow e^{\rho\sigma_z/2}u \iff U^\mu \rightarrow \Lambda^\mu{}_\nu(\hat{z}, \rho) U^\nu, \quad \tanh \rho = v/c.} \quad (16)$$

Phase rotation. For $u \rightarrow e^{i\theta\sigma_z}u$, $\sigma_z^2 = \mathbf{1}$:

$$e^{i\theta\sigma_z} = \cos \theta \mathbf{1} + i \sin \theta \sigma_z.$$

Acting on $u = (a, b)^T$: $u' = (e^{i\theta}a, e^{-i\theta}b)$. Compute components:

$$U'^0 = \frac{1}{2}(|a|^2 + |b|^2) = U^0,$$

$$U'^z = U^z,$$

$$U'^x = \text{Re}(e^{-i\theta}a^* \cdot e^{-i\theta}b) = \text{Re}(e^{-2i\theta}a^*b) = \cos(2\theta)U^x + \sin(2\theta)U^y,$$

$$U'^y = -\sin(2\theta)U^x + \cos(2\theta)U^y.$$

This is a spatial *rotation* about $\hat{z}$ by angle 2θ (note the factor of two: spinor rotations are double-cover of $SO(3)$). Hence

$$\boxed{u \rightarrow e^{i\theta\sigma_z}u \iff U^\mu \rightarrow R^\mu{}_\nu(\hat{z}, 2\theta) U^\nu.} \quad (17)$$

(e) Does a 2-spinor satisfy the massless KG equation?

A null 4-vector $K^\mu = \frac{1}{2}u^\dagger\sigma^\mu u$ satisfies $K_\mu K^\mu = 0$, exactly the dispersion of a massless plane wave. So if $u(x) \propto e^{ik \cdot x}$ with $k_\mu k^\mu = 0$, the bilinear $u^\dagger\sigma^\mu u$ is the 4-momentum of a massless particle, and each component of u obeys $\partial_\mu\partial^\mu u_a = 0$, i.e. the massless Klein–Gordon equation. The natural first-order equation is the *Weyl equation* $\bar{\sigma}^\mu\partial_\mu u = 0$; squaring it gives the massless KG. Hence

Yes: each component of a massless 2-spinor solution satisfies the massless KG equation.	(18)
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